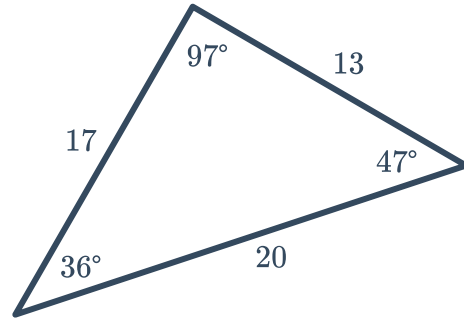


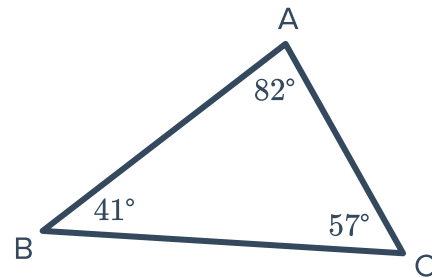
Inequalities in one triangle



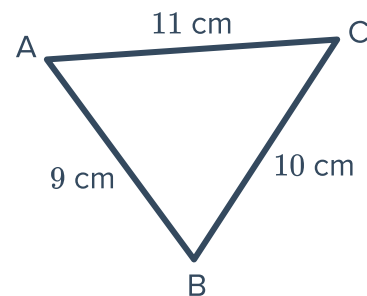
- 1 Consider the given triangle and think about the longest and shortest sides.
 - a Find the sum of the interior angles in the triangle.
 - b Which side appears opposite the largest angle?
 - c Which side appears opposite the smallest angle?



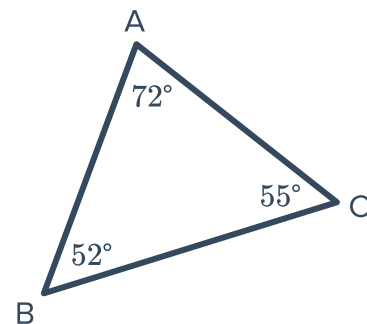
- 2 Consider the given triangle.
Which side is the longest side of the triangle?



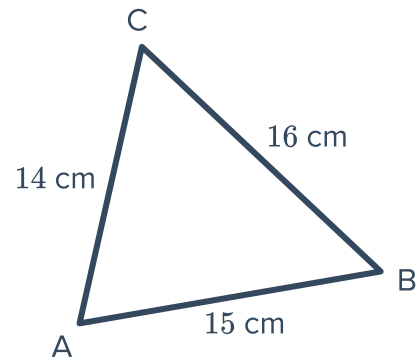
- 3 Consider the given triangle.
Which angle is the largest angle of the triangle?



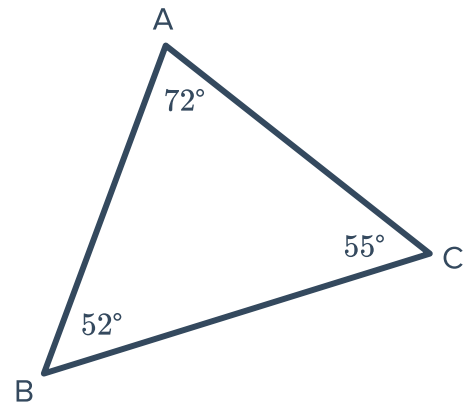
- 4 Consider the given triangle.
Which side is the shortest side of the triangle?



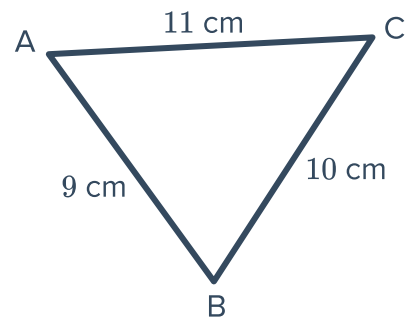
- 5 Consider the given triangle.
Which angle is the smallest angle of the triangle?



- 6 Consider the given triangle.
- List the sides of the triangle from smallest to largest.
 - Write an inequality relating the lengths of the three sides of the triangle.



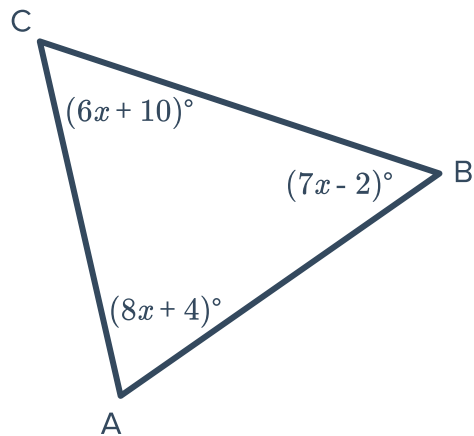
- 7 Consider the given triangle.
- List the angles of the triangle from smallest to largest.
 - Write an inequality relating the measure of the three angles of the triangle.



8 Consider the given triangle.

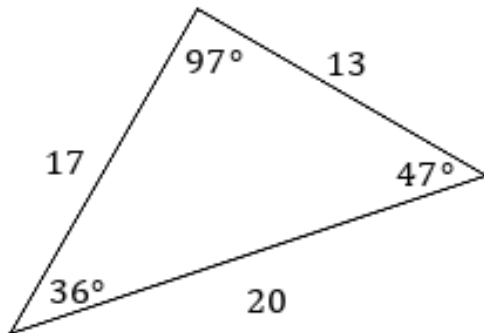


- a Solve for x .
- b Find $m\angle CAB$.
- c Find $m\angle ABC$.
- d Find $m\angle BCA$.
- e Hence, which side is the longest side of the triangle?



Additional questions

- 9 Consider the triangle in the given figure. Describe the relationship between the sizes of the angle and opposite side compared to the other opposite side-angle pairs.



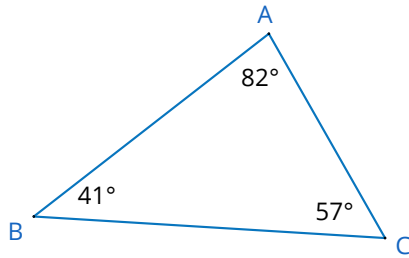
10 Complete the following theorems:

- a If one side of a triangle is longer than another side, then the angle opposite the longer side is...
- b If one angle of a triangle is larger than another angle, then the side opposite the larger angle is...

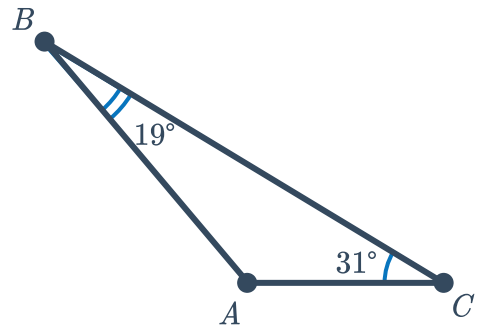
11 For each of the following triangles, order the sides from shortest to longest.



a



b



c $\triangle ABC$ where,

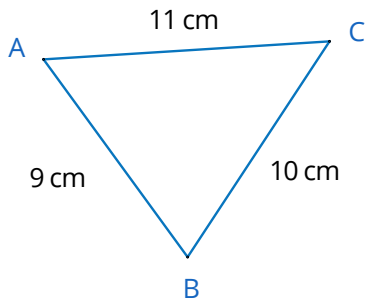
- $m\angle A = 37^\circ$
- $m\angle B = 90^\circ$
- $m\angle C = 53^\circ$

d $\triangle ABC$ where,

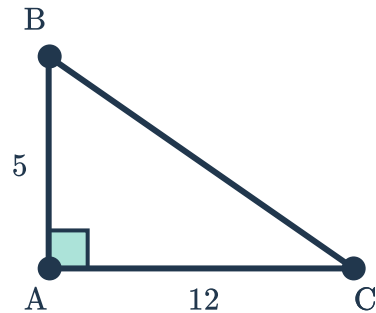
- $m\angle A = 115^\circ$
- $m\angle B = 32^\circ$

12 For each of the following triangles, order the angles from smallest to largest.

a



b



c $\triangle ABC$ where,

- $AB = 7$
- $AC = 3.2$
- $BC = 4.8$

d $\triangle ABC$ where,

- $AB = 50$
- $AC = 62$
- $BC = 70$

13 Draw the following triangles using the given measures.



a Side lengths

- 12
- 6
- 8

Angle measures

- 24°
- 120°
- 36°

b Side lengths

- 8
- 6
- 10

Angle measures

- 53°
- 37°

14 Given that the hypotenuse is the longest side in a right triangle, explain why a right triangle must have one right angle and two acute angles, without using the triangle angle sum theorem.